Chapter 5: Real-time Estimation of Spatially Correlated 3-D Drag Coefficient Matrix

5.1 Introduction

Real-time estimation of input gain enabled adaptive self-tuning control of the magnetic microprobe in aqueous solution. In Chapter 4, we discussed that the drag coefficient can be characterized by two principal directions, perpendicular and parallel to the surface. Thus, we can estimate the drag coefficient parallel and vertical drag coefficient respectively. This statement is true when the surface normal is overlapped with the z-direction in the measurement coordinate. In an unstructured environment, i.e., incline flat surface and sphere surface, there is no theoretical model for the wall effect. The three separate drag coefficients will turn into a 3×3 drag coefficient matrix depending on the orientation of the surface normal.

In Chapter 4, we used the variance to predict the estimated input gain for the three axes. In this Chapter, the first objective is to estimate the orientation of the surface. We utilize the covariance matrix of the tracking error to predict and estimate the orientation of the surface through real-time Principal Component Analysis [60]. Since the covariance matrix is positive definite symmetric matrix, it can be expressed in the form of diagonalized eigenvalues and eigenvectors [66]. The eigenvectors are characterized by three angles, and we derived a relationship that we can real-time update the eigenvector and the eigenvalue.

With the information of the eigenvector, we can transform the system from the measurement coordinate to the local coordinate by multiplying the eigenvector matrix. The 3×3 input matrix of the probe in the measurement coordinate is diagonalized in the local coordinate. The control law and augmented estimator can be derived similarly to the method discussed in Chapter 4. It is worth noting that the estimation of the orientation of the surface is included in the augmented estimator. By implementing the control law and Kalman-filter based augmented estimator, we can achieve zero-mean tracking error, minimum variance control, and surface orientation estimation.

5.2 3x3 Drag Coefficient Matrix in Unknown Environment

In Chapter 2.7.1, it is stated that the drag coefficient varies with the distance between the surface and the center of the probe. It is also stated that the drag coefficient can be characterized by two principal directions, perpendicular and parallel to the surface. This statement is valid when the surface normal direction is overlapped with the z-direction in the measurement coordinate. In an unstructured environment, i.e., incline flat surface and sphere surface, there is no theoretical model for the wall effect. Specifically, in the case that the wall effect cannot be ignored, the drag coefficient needs to be extended to a 3×3 drag coefficient matrix which not only depends on the dimensionless variable R/h, i.e., the ratio between the radius of and the distance of the bead from the wall, but also on the orientation of the surface as the two correction coefficients for the wall effect are not equal, i.e., $C_{\perp}(R/h) \neq C_{\parallel}(R/h)$. Therefore, it is important to estimate the orientation of the surface.

5.3 Estimate Drag Coefficient Matrix through Principal Component Analysis

In Chapter 4, we used (4.8) the variance of the tracking error to predict the estimated input gain for the three axes. In the unknown environment, we extended the variance of the tracking error to the covariance matrix of the tracking error to predict and estimate the surface normal and the 3×3 input gain matrix through real-time Principal Component Analysis [60]. The covariance matrix of the tracking error $\mathbf{R}$ or the input gain matrix $\mathbf{A}$ is a 3×3 positive definite symmetric matrix. They can both be expressed in the form of eigenvalues and eigenvectors,

$$\mathbf{R} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{\mathbf{T}} \tag{5.1}$$

wherein $\mathbf{V} = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3]$, $\boldsymbol{\Lambda} = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$, and $\mathbf{V}\mathbf{V}^T = \mathbf{I}$. It is worth noting that eigenvector $\mathbf{V}$ is characterized by three parameters, i.e., $\mathbf{V}$= $\mathbf{V}(\theta_1, \theta_2, \theta_3)$. With respect to the coordinate system $\mathbf{V}$, ${}^{\mathbf{V}}\mathbf{R} = \mathbf{V}^T\mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^T\mathbf{V} = \boldsymbol{\Lambda}$. The relationship between the covariance matrix and the input gain matrix is,

$$\mathbf{R}_{\mathbf{V}_{\delta\mathbf{x}}} = \left\{2 + \frac{1}{(1-\lambda^2)}\right\}(4k_bT){}^{\mathbf{V}}\mathbf{A}(z) = k_{RA}{}^{\mathbf{V}}\mathbf{A}(z) \tag{5.2}$$

We use a single-pole Infinite Impulse Response (IIR) filter to real-time estimate the covariance matrix of the measured tracking error.

$$\widehat{\mathbf{R}}[k+1] = (1 - a_{cov})\widehat{\mathbf{R}}[k] + a_{cov}\delta\mathbf{x}[k]\delta\mathbf{x}[k]^T \tag{5.3}$$

where $(1 - a_{cov})$ is the pole location which determines the effective length of calculating the covariance matrix of the measured tracking error, and $\delta\mathbf{x}[k]$ is the newly arrived 3-

dimensional tracking error. Define $\delta\mathbf{x}_{RA}[k] = \delta\mathbf{x}[k]/\sqrt{k_{RA}}$. We use the same method for the real-time estimation of input gain matrix $\widehat{\mathbf{A}}[k]$.

$$\widehat{\mathbf{A}}[k+1] = (1 - a_{cov})\widehat{\mathbf{A}}[k] + a_{cov}\delta\mathbf{x}_{RA}[k]\delta\mathbf{x}_{RA}[k]^T \tag{5.4}$$

Rearrange (5.4), the changing of the input gain matrix is

$$\delta\mathbf{A}[k] = \widehat{\mathbf{A}}[k+1] - \widehat{\mathbf{A}}[k] = a_{cov}\{\delta\mathbf{x}_{RA}[k]\delta\mathbf{x}_{RA}[k]^T - \widehat{\mathbf{A}}[k]\} \tag{5.5}$$

Since $\widehat{\mathbf{A}}[k] = \mathbf{V}[\mathrm{k}]\mathbf{\Lambda}[\mathrm{k}]\mathbf{V}^T[k]$, equation (5.5) can be rewritten as

$$\delta\mathbf{A}[k] = a_{cov}\widehat{\mathbf{V}}[k]\{\widehat{\mathbf{V}}^T[k]\delta\mathbf{x}_{RA}[k]\delta\mathbf{x}_{RA}[k]^T\widehat{\mathbf{V}}[k] - \widehat{\mathbf{\Lambda}}[k]\}\widehat{\mathbf{V}}^T[k] \tag{5.6}$$

Transform the system from the measurement coordinate to the coordinate system $\mathbf{V}[k]$, where ${}^{\mathbf{V}}\delta\mathbf{x}_{RA}[k] = \mathbf{V}^T[k]\delta\mathbf{x}_{RA}[k]$.

$${}^{\mathbf{V}}\delta\mathbf{A}[k] = a_{cov}\{{}^{\mathbf{V}}\delta\mathbf{x}_{RA}[k]{}^{\mathbf{V}}\delta\mathbf{x}_{RA}[k]^T - \widehat{\mathbf{\Lambda}}[k]\} \tag{5.7}$$

According to the definition of the changing of the input gain, (5.7) can be rewritten as

$${}^{\mathbf{V}}\delta\mathbf{A}[k] = \begin{bmatrix} {}^{\mathbf{V}}\delta x_{RA1}^2[k] - \hat{\lambda}_1[k] & {}^{\mathbf{V}}\delta x_{RA1}[k]{}^{\mathbf{V}}\delta x_{RA2}[k] & {}^{\mathbf{V}}\delta x_{RA3}[k]{}^{\mathbf{V}}\delta x_{RA1}[k] \\ {}^{\mathbf{V}}\delta x_{RA1}[k]{}^{\mathbf{V}}\delta x_{RA2}[k] & {}^{\mathbf{V}}\delta x_{RA2}^2[k] - \hat{\lambda}_2[k] & {}^{\mathbf{V}}\delta x_{RA2}[k]{}^{\mathbf{V}}\delta x_{RA3}[k] \\ {}^{\mathbf{V}}\delta x_{RA3}[k]{}^{\mathbf{V}}\delta x_{RA1}[k] & {}^{\mathbf{V}}\delta x_{RA2}[k]{}^{\mathbf{V}}\delta x_{RA3}[k] & {}^{\mathbf{V}}\delta x_{RA3}^2[k] - \hat{\lambda}_3[k] \end{bmatrix} \tag{5.8}$$

The eigenvector of the input gain matrix $\mathbf{A}[k]$ can be characterized by three angles, i.e., θ_1, θ_2, and θ_3. We assumed that the variation of the angles of the eigenvector is small, i.e., $\delta\theta_1$, $\delta\theta_2$, and $\delta\theta_3$. The changing of the eigenvalue is defined as $\delta\lambda_1$, $\delta\lambda_2$, and $\delta\lambda_3$

$${}^{\mathbf{V}}\delta\mathbf{A}[k] = \begin{bmatrix} \delta\lambda_1 & \delta\theta_3(\lambda_1 - \lambda_2) & \delta\theta_2(\lambda_3 - \lambda_1) \\ \delta\theta_3(\lambda_1 - \lambda_2) & \delta\lambda_2 & \delta\theta_1(\lambda_2 - \lambda_3) \\ \delta\theta_2(\lambda_3 - \lambda_1) & \delta\theta_1(\lambda_2 - \lambda_3) & \delta\lambda_3 \end{bmatrix} \tag{5.9}$$

We define the changing of the eigenvalue and the variation of the angle of the eigenvector as

$$\begin{cases} \delta\mathbf{\Lambda}[k] = \begin{bmatrix} \delta\lambda_1[k] \\ \delta\lambda_2[k] \\ \delta\lambda_3[k] \end{bmatrix} \\ \delta\boldsymbol{\theta}_{\mathbf{\Lambda}}[k] = \begin{bmatrix} (\hat{\lambda}_1[k] - \hat{\lambda}_2[k])\delta\theta_3[k] \\ (\hat{\lambda}_3[k] - \hat{\lambda}_1[k])\delta\theta_2[k] \\ (\hat{\lambda}_2[k] - \hat{\lambda}_3[k])\delta\theta_1[k] \end{bmatrix} \end{cases} \tag{5.10}$$

The update of the eigenvalue and the angle of the eigenvector are

$$\begin{cases} \mathbf{\Lambda}[k+1] = \mathbf{\Lambda}[k] + \delta\mathbf{\Lambda}[k] \\ \boldsymbol{\theta}_{\mathbf{\Lambda}}[k+1] = \boldsymbol{\theta}_{\mathbf{\Lambda}}[k] + \delta\boldsymbol{\theta}_{\mathbf{\Lambda}}[k] \end{cases} \tag{5.11}$$

The prediction based on the measured tracking error in the local coordinate is

$$\begin{cases} \mathbf{\Lambda}[\mathrm{k}+1] = (1 - a_{cov})\mathbf{\Lambda}[\mathrm{k}] + a_{cov} \begin{bmatrix} {}^{\mathbf{V}}\delta x_{RA1}^2[k] \\ {}^{\mathbf{V}}\delta x_{RA2}^2[k] \\ {}^{\mathbf{V}}\delta x_{RA3}^2[k] \end{bmatrix} \\ \boldsymbol{\theta}_{\mathbf{\Lambda}}[k+1] = \boldsymbol{\theta}_{\mathbf{\Lambda}}[k] + a_{cov} \begin{bmatrix} {}^{\mathbf{V}}\delta x_{RA2}[k] \cdot {}^{\mathbf{V}}\delta x_{RA3}[k] \\ {}^{\mathbf{V}}\delta x_{RA3}[k] \cdot {}^{\mathbf{V}}\delta x_{RA1}[k] \\ {}^{\mathbf{V}}\delta x_{RA1}[k] \cdot {}^{\mathbf{V}}\delta x_{RA2}[k] \end{bmatrix} \end{cases} \tag{5.12}$$

The update for the eigenvector matrix is

$$\hat{\mathbf{V}}[k+1] = \hat{\mathbf{V}}[k] \begin{bmatrix} 1 & -\delta\theta_3 & \delta\theta_2 \\ \delta\theta_3 & 1 & -\delta\theta_1 \\ -\delta\theta_2 & \delta\theta_1 & 1 \end{bmatrix} \tag{5.13}$$

wherein the variance of the three angles of the eigenvector is

$$\begin{bmatrix} \delta\theta_1 \\ \delta\theta_2 \\ \delta\theta_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{(\hat{\lambda}_2[k] - \hat{\lambda}_3[k])} & 0 & 0 \\ 0 & \frac{1}{(\hat{\lambda}_3[k] - \hat{\lambda}_1[k])} & 0 \\ 0 & 0 & \frac{1}{(\hat{\lambda}_1[k] - \hat{\lambda}_2[k])} \end{bmatrix} \boldsymbol{\theta}_{\mathbf{\Lambda}}[\mathrm{k}] \tag{5.14}$$

5.4 Control Law and Estimator Design in Local Coordinate

The probe dynamic considering random environment using the zero-order hold transformation can be expressed by the difference equation

$$\mathbf{x}[k+1] = \mathbf{x}[k] + \mathbf{A}[k](\mathbf{f}_d[k] + \mathbf{d}[k] + \mathbf{F}_T[k]) \tag{5.15}$$

where $\mathbf{x}[k] = [x[k] \quad y[k] \quad z[k]]^T$, $\mathbf{A}[k]$ is the 3×3 input gain matrix considering the random environment, $\mathbf{d}[k]$ is the lumped of the modeling error term and the disturbance vector, and $\mathbf{F}_T[k]$ is the random thermal force vector. To simplify the task, it is worth noting that the 3×3 input gain matrix $\mathbf{A}[k]$ becomes diagonal matrix in the local coordinate $\mathbf{V}$. We define the coordinate transformation matrix from the measurement coordinate to the local coordinate as ${}^{\mathbf{V}}_{\mathbf{M}}\mathbf{R}[k] = \mathbf{V}^T[k]$. By left multiplying ${}^{\mathbf{V}}_{\mathbf{M}}\mathbf{R}[k]$ to (5.15), the probe dynamic is the local coordinate is expressed as,

$${}^{\mathbf{V}}\mathbf{x}[k+1] = {}^{\mathbf{V}}\mathbf{x}[k] + {}^{\mathbf{V}}\mathbf{\Lambda}[k]\left({}^{\mathbf{V}}\mathbf{f_d}[k] + {}^{\mathbf{V}}\mathbf{d}[k] + {}^{\mathbf{V}}\mathbf{F}_T[k]\right) \tag{5.16}$$

Let the control objective be ${}^{\mathbf{V}}\delta\mathbf{x}[k+1] = \lambda_c \cdot {}^{\mathbf{V}}\delta\mathbf{x}[k]$, where ${}^{\mathbf{V}}\delta\mathbf{x}[k] = {}^{\mathbf{V}}\mathbf{x_d}[k] - {}^{\mathbf{V}}\mathbf{x}[k]$, and ${}^{\mathbf{V}}\mathbf{x_d}[k]$ is the desired motion trajectory in the local coordinate, comparing (5.16) with the desired motion, the force required to meet the control objective in the local coordinate can be obtained by using the estimated tracking error, input gain matrix, and disturbance,

$${}^{\mathbf{V}}\mathbf{f_d}[k] = {}^{\mathbf{V}}\widehat{\mathbf{\Lambda}}^{-1}\left\{{}^{\mathbf{V}}\mathbf{x_d}[k+1] - {}^{\mathbf{V}}\mathbf{x_d}[k] + (1-\lambda_c){}^{\mathbf{V}}\delta\mathbf{x}[k]\right\} - {}^{\mathbf{V}}\mathbf{D}[k] \tag{5.17}$$

Substituting (5.17) into (5.16) and comparing the resulting motion with the desired motion trajectory, the dynamics of the tracking error can be derived,

$${}^{\mathbf{V}}\delta\mathbf{x}[k+1] = \lambda_c \cdot {}^{\mathbf{V}}\delta\mathbf{x}[k] + \boldsymbol{\varepsilon}[k] \tag{5.18}$$

where $\boldsymbol{\varepsilon}[k]$ is a zero-mean random variable related to thermal force, estimation error, and λ_c, i.e.,

$$\boldsymbol{\varepsilon}[k] = (1-\lambda_c)\,{}^{\mathbf{V}}\mathbf{e}_{\delta x}[k] - {}^{\mathbf{V}}\mathbf{E}_{\mathbf{A}}[k]\,{}^{\mathbf{V}}\widehat{\boldsymbol{\Lambda}}^{-1}[k]\left\{{}^{\mathbf{V}}\mathbf{x}_{\mathbf{d}}[k+1] - {}^{\mathbf{V}}\mathbf{x}_{\mathbf{d}}[k] + (1-\lambda_c)\,{}^{\mathbf{V}}\delta\hat{\mathbf{x}}[k]\right\} \\ -{}^{\mathbf{V}}\boldsymbol{\Lambda}[k]\left({}^{\mathbf{V}}\mathbf{d}[k] - {}^{\mathbf{V}}\hat{\mathbf{d}}[k]\right) - {}^{\mathbf{V}}\boldsymbol{\Lambda}[k]\,{}^{\mathbf{V}}\mathbf{F}_T[k] \tag{5.19}$$

Kalman filter-based augmented estimator is designed to estimate time-variant input gain matrix in the local coordinate in real-time to render adaptive self-tuning control when propelling the probe in aqueous solutions in random environment. Based on the tracking error (5.19), the Kalman filter-based augmented estimator can be derived as,

Let ${}^{\mathbf{V}}\delta\mathbf{x}_1[k] = {}^{\mathbf{V}}\delta\mathbf{x}_m[k], {}^{\mathbf{V}}\delta\mathbf{x}_2[k] = {}^{\mathbf{V}}\delta\mathbf{x}_m[k+1], {}^{\mathbf{V}}\delta\mathbf{x}_3[k] = {}^{\mathbf{V}}\delta\mathbf{x}_m[k+2]$, ${}^{\mathbf{V}}\mathbf{d}[k+1] = {}^{\mathbf{V}}\mathbf{d}[k] + {}^{\mathbf{V}}\delta\mathbf{d}[k], {}^{\mathbf{V}}\delta\mathbf{d}[k+1] = {}^{\mathbf{V}}\delta\mathbf{d}[k]$, ${}^{\mathbf{V}}\boldsymbol{\Lambda}[\mathrm{k}+1] = {}^{\mathbf{V}}\boldsymbol{\Lambda}[\mathrm{k}] + \delta^{\mathbf{V}}\boldsymbol{\Lambda}[\mathrm{k}], \boldsymbol{\theta}_{\boldsymbol{\Lambda}}[\mathrm{k}+1] = \boldsymbol{\theta}_{\boldsymbol{\Lambda}}[\mathrm{k}] + \delta\boldsymbol{\theta}_{\boldsymbol{\Lambda}}[\mathrm{k}], \delta^{\mathbf{V}}\boldsymbol{\Lambda}[\mathrm{k}+1] = \delta^{\mathbf{V}}\boldsymbol{\Lambda}[\mathrm{k}]$, *and* $\delta\boldsymbol{\theta}_{\boldsymbol{\Lambda}}[\mathrm{k}+1] = \delta\boldsymbol{\theta}_{\boldsymbol{\Lambda}}[\mathrm{k}]$

The prediction of the state is

$$\underbrace{\begin{bmatrix} {}^{\mathbf{V}}\delta\widehat{\boldsymbol{x}}_1^f[k+1] \\ {}^{\mathbf{V}}\delta\widehat{\boldsymbol{x}}_2^f[k+1] \\ {}^{\mathbf{V}}\delta\widehat{\boldsymbol{x}}_3^f[k+1] \\ {}^{\mathbf{V}}\widehat{\boldsymbol{d}}^f[k+1] \\ {}^{\mathbf{V}}\delta\widehat{\boldsymbol{d}}^f[k+1] \\ {}^{\mathbf{V}}\widehat{\boldsymbol{\Lambda}}^f[k+1] \\ \delta^{\mathbf{V}}\widehat{\boldsymbol{\Lambda}}^f[k+1] \\ \widehat{\boldsymbol{\theta}}_{\boldsymbol{\Lambda}}^f[k+1] \\ \delta\widehat{\boldsymbol{\theta}}_{\boldsymbol{\Lambda}}^f[k+1] \end{bmatrix}}_{\widehat{X}^f[k+1]} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \lambda_c & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \underbrace{\begin{bmatrix} {}^{\mathbf{V}}\delta\widehat{\boldsymbol{x}}_1[k] \\ {}^{\mathbf{V}}\delta\widehat{\boldsymbol{x}}_2[k] \\ {}^{\mathbf{V}}\delta\widehat{\boldsymbol{x}}_3[k] \\ {}^{\mathbf{V}}\widehat{\boldsymbol{d}}[k] \\ {}^{\mathbf{V}}\delta\widehat{\boldsymbol{d}}[k] \\ {}^{\mathbf{V}}\widehat{\boldsymbol{\Lambda}}[k] \\ \delta^{\mathbf{V}}\widehat{\boldsymbol{\Lambda}}[k] \\ \widehat{\boldsymbol{\theta}}_{\boldsymbol{\Lambda}}[k] \\ \delta\widehat{\boldsymbol{\theta}}_{\boldsymbol{\Lambda}}[k] \end{bmatrix}}_{\widehat{X}[k]} \tag{5.20}$$

The output of the estimator is ${}^{\mathbf{V}}\boldsymbol{y}[k] = \begin{bmatrix} {}^{\mathbf{V}}\delta\boldsymbol{x}_m[k] \\ {}^{\mathbf{V}}\boldsymbol{\Lambda}[k] \\ \boldsymbol{\theta}_{\boldsymbol{\Lambda}}[k] \end{bmatrix}, {}^{\mathbf{V}}\widehat{\boldsymbol{y}}[\boldsymbol{k}][k] = \begin{bmatrix} {}^{\mathbf{V}}\delta\widehat{\boldsymbol{x}}_m[k] \\ {}^{\mathbf{V}}\widehat{\boldsymbol{\Lambda}}[k] \\ \widehat{\boldsymbol{\theta}}_{\boldsymbol{\Lambda}}[k] \end{bmatrix}$

The estimation error dynamic is

$$\underbrace{\begin{bmatrix} \boldsymbol{e}_{\mathrm{v}_{\delta \mathbf{x}_1}}[k+1] \\ \boldsymbol{e}_{\mathrm{v}_{\delta \mathbf{x}_2}}[k+1] \\ \boldsymbol{e}_{\mathrm{v}_{\delta \mathbf{x}_3}}[k+1] \\ \boldsymbol{e}_{\mathrm{v}_{\mathbf{d}}}[k+1] \\ \boldsymbol{e}_{\mathrm{v}_{\delta \mathbf{d}}}[k+1] \\ \boldsymbol{e}_{\mathrm{v}_{\boldsymbol{\Lambda}}}[k+1] \\ \boldsymbol{e}_{\delta \mathrm{v}_{\boldsymbol{\Lambda}}}[k+1] \\ \boldsymbol{e}_{\boldsymbol{\theta}_{\Lambda}}[k+1] \\ \boldsymbol{e}_{\delta \boldsymbol{\theta}_{\Lambda}}[k+1] \end{bmatrix}}_{\boldsymbol{E}[k+1]} = \underbrace{\begin{bmatrix} 0 & \mathbf{I_{3x3}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \mathbf{I_{3x3}} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \mathbf{I_{3x3}} & -{}^{\mathrm{V}}\widehat{\boldsymbol{\Lambda}}[k] & 0 & -\mathbf{G}_{\boldsymbol{\Lambda}}[k] & 0 & -\mathbf{G}_{\boldsymbol{\theta}_{\Lambda}}[k] & 0 \\ 0 & 0 & 0 & \mathbf{I_{3x3}} & \mathbf{I_{3x3}} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \mathbf{I_{3x3}} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \mathbf{I_{3x3}} & \mathbf{I_{3x3}} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \mathbf{I_{3x3}} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \mathbf{I_{3x3}} & \mathbf{I_{3x3}} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \mathbf{I_{3x3}} \end{bmatrix}}_{\boldsymbol{F}[k]} \underbrace{\begin{bmatrix} \boldsymbol{e}_{\mathrm{v}_{\delta \mathbf{x}_1}}[k] \\ \boldsymbol{e}_{\mathrm{v}_{\delta \mathbf{x}_2}}[k] \\ \boldsymbol{e}_{\mathrm{v}_{\delta \mathbf{x}_3}}[k] \\ \boldsymbol{e}_{\mathrm{v}_{\mathbf{d}}}[k] \\ \boldsymbol{e}_{\mathrm{v}_{\delta \mathbf{d}}}[k] \\ \boldsymbol{e}_{\mathrm{v}_{\boldsymbol{\Lambda}}}[k] \\ \boldsymbol{e}_{\delta \mathrm{v}_{\boldsymbol{\Lambda}}}[k] \\ \boldsymbol{e}_{\boldsymbol{\theta}_{\Lambda}}[k] \\ \boldsymbol{e}_{\delta \boldsymbol{\theta}_{\Lambda}}[k] \end{bmatrix}}_{\boldsymbol{E}[k]} \quad (5.21)$$

where $\mathbf{G}_{\boldsymbol{\Lambda}}[k]$ and $\mathbf{G}_{\boldsymbol{\theta}_{\Lambda}}[k]$ are matrix of the component of ${}^{\mathrm{V}}\boldsymbol{\Lambda}[k]$.

The prediction of the error covariance is

$$\mathbf{P}^{f}[k] = \mathbf{F}[k]\mathbf{P}[k-1]\mathbf{F}^{T}[k] + \mathbf{Q}[k-1] \quad (5.22)$$

where $\mathbf{P}[k-1]$ is the error covariance matrix, and $\mathbf{Q}[k-1]$ is the variance of the process noise.

The estimator gain matrix is

$$\mathbf{K}[k] = \mathbf{P}^{f}[k]\mathbf{H}^{T}(\mathbf{H}\mathbf{P}^{f}[k]\mathbf{H}^{T} + \mathbf{R}_{k})^{-1} \quad (5.23)$$

where $\mathbf{R}_k$ is the variance of the measurement noise, and define

$$\mathbf{H} = \begin{bmatrix} \mathbf{I_{3x3}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \mathbf{I_{3x3}} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \mathbf{I_{3x3}} & 0 \end{bmatrix} \quad (5.24)$$

Update the estimation through the estimator gain

$$\widehat{\boldsymbol{X}}[k+1] = \widehat{\boldsymbol{X}}^{f}[k+1] + \mathbf{K}[k](\boldsymbol{y}[k] - \widehat{\boldsymbol{y}}[k]) \quad (5.25)$$

Update the error covariance

$$\mathbf{P}[k] = \frac{1}{\lambda_f}[\boldsymbol{I} - \mathbf{K}[k]\mathbf{H}]\mathbf{P}^{f}[k] \quad (5.26)$$

where λ_f is the forgetting factor which gives exponentially less weight to older error samples.

5.5 Experimental Verification of the 3-D Drag Coefficient Matrix Estimation

Experiments are designed to verify the 3-D drag coefficient matrix estimation and the control law derived in the local coordinate. In the first experiment, instead of placing the coverslip perpendicular to the z axis of the measurement coordinate, we rotated the coverslip 45° by the x axis of the measurement coordinate. Therefore, the scanning microprobe is immersed in water and subjected to the coverslip which can be viewed as a tilted wall. The probe started from the initial position of $(0\ \mu m, 0\ \mu m, 0\ \mu m)$. The motion of the probe is a 11.5-micrometer ramp to the negative direction of w, which is approaching the tilted wall in the normal direction. The speed is controlled as 5 um/sec. The other two directions are stabilized at the original point. The probe is stabilized and controlled by the control law (5.17). We used the augmented Kalman-filter-based estimator (5.20) to (5.26) to estimate the direction of the surface normal, the 3×3 input gain matrix, and the disturbance. It is worth noting that the 3×3 input gain matrix is decoupled when we transform to the local coordinate.

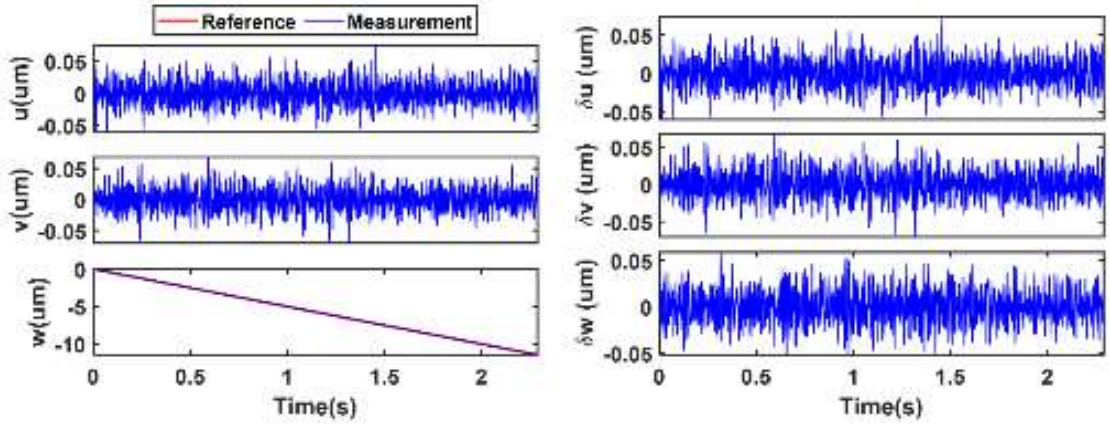

Fig. 38. The left panel shows the resulting motion in the local coordinate, and the right panel shows the corresponding tracking error

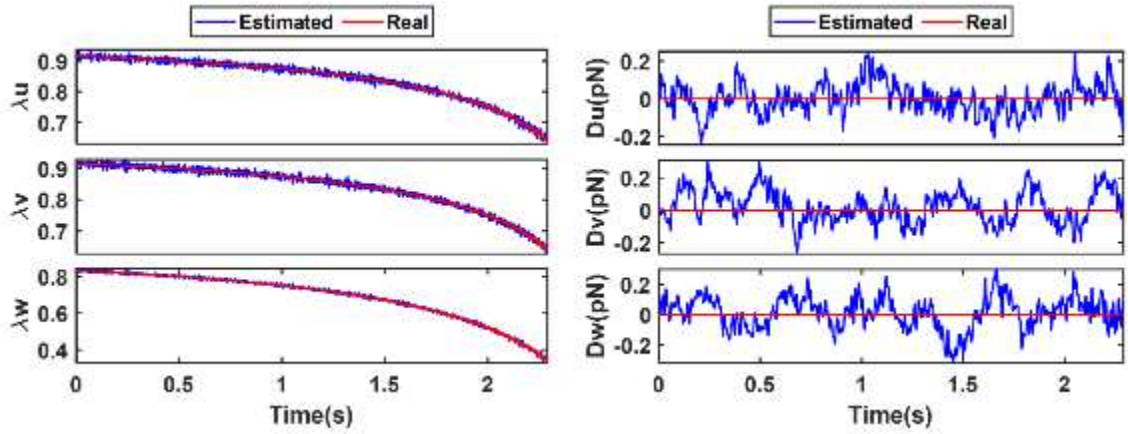

Fig. 39. The left panel shows the real-time estimation of the normalized input gain in the local coordinate, and the right panel shows the estimated disturbance.

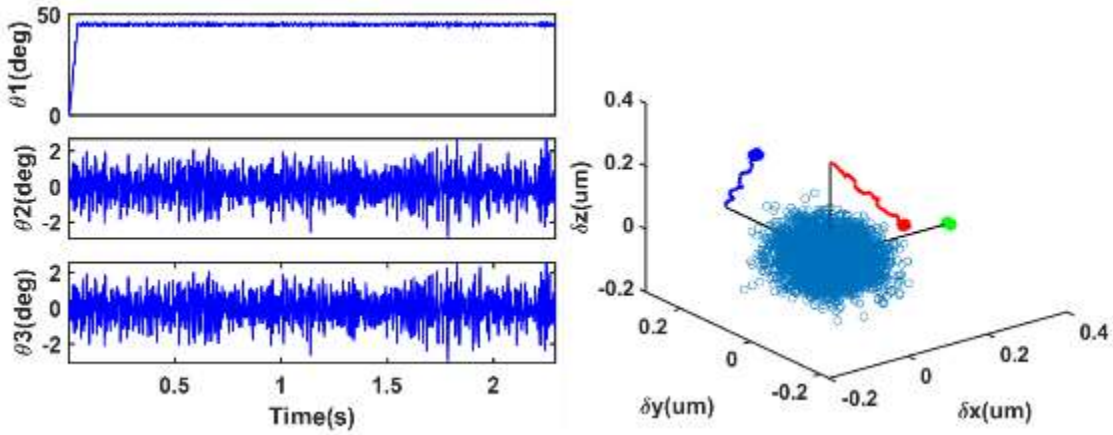

Fig. 40. The left panel shows the real-time estimation of the orientation of the surface normal, and the black line of the right panel is the positive direction of the measurement coordinate, while the dark blue, red, and green curve are the estimation result of the orientation of the surface normal in the measurement coordinate.

The left panel of Fig. 38 shows the result of motion control with real-time estimation and adaptive self-tuning control in the local coordinate. The right panel of Fig. 38 shows the corresponding tracking error. The mean of the tracking error is zero for all three axes. The left panel of Fig. 39 shows the real-time estimation of the normalized input gain which is normalized by the nominal input gain in the local coordinate. The real-time estimated input gain followed closely with the real input gain. The right panel of Fig. 39 shows the estimated disturbance. The estimated disturbance is not zero because of the thermal force and measurement noise. The left panel of Fig. 40 shows the real-time estimated orientation of the surface normal. The three angles, i.e., θ_1, θ_2 and θ_3, represent the rotation with respect to x, y, and z. The estimation result of θ_1 converged to 45° which matched the design of the experiment. The black line of the right panel is the

positive direction of the measurement coordinate. The light blue circles are the measured tracking error that are used to estimate the orientation of the surface normal. The green, dark blue, and red curves are the estimation result of the orientation of the surface normal in the measurement coordinate.

We repeated the experiment by rotating the coverslip 15° in the y axis of the measurement coordinate. The left panel of Fig. 41 shows the result of motion control with real-time estimation and adaptive time variant control in the local coordinate. The right panel of Fig. 41 shows the corresponding tracking error. We achieved zero-mean tracking error for all three axes. The result is similar to the first experiment since they are both in the local coordinate. The left panel of Fig. 42 shows the real-time estimation of the normalized input gain which is normalized by the nominal input gain in the local coordinate. The real-time estimated input gain followed closely with the real input gain. The right panel of Fig. 42 shows the estimated disturbance which is not zero because of the thermal force and measurement noise. The left panel of Fig. 43 shows the real-time estimated orientation of the surface normal. The estimation result of θ_2 converged to 15° which matched the design of the experiment. The black line of the right panel is the positive direction of the measurement coordinate. The light blue circles are the measured tracking error that are used to estimate the orientation of the surface normal. The green, dark blue, and red curves are the estimation result of the orientation of the surface normal in the measurement coordinate.

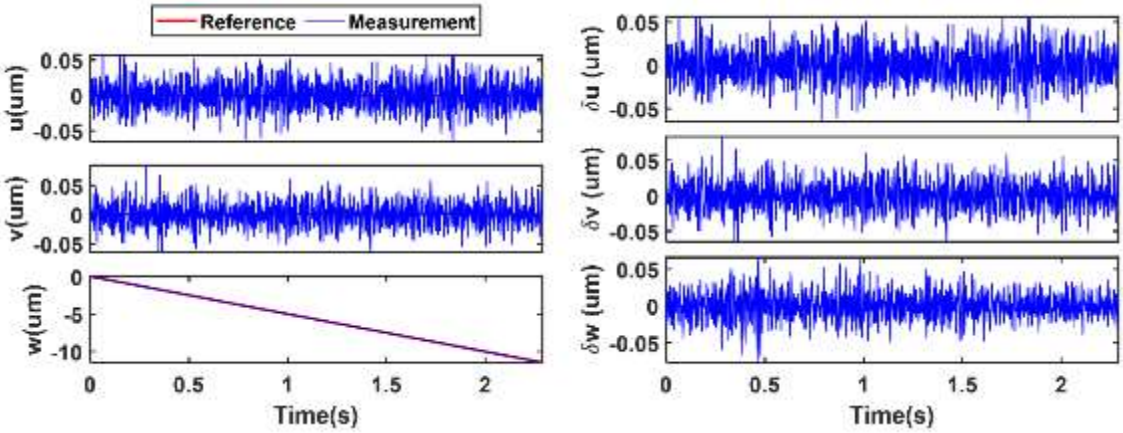

Fig. 41. The left panel shows the resulting motion in the local coordinate, and the right panel shows the corresponding tracking error

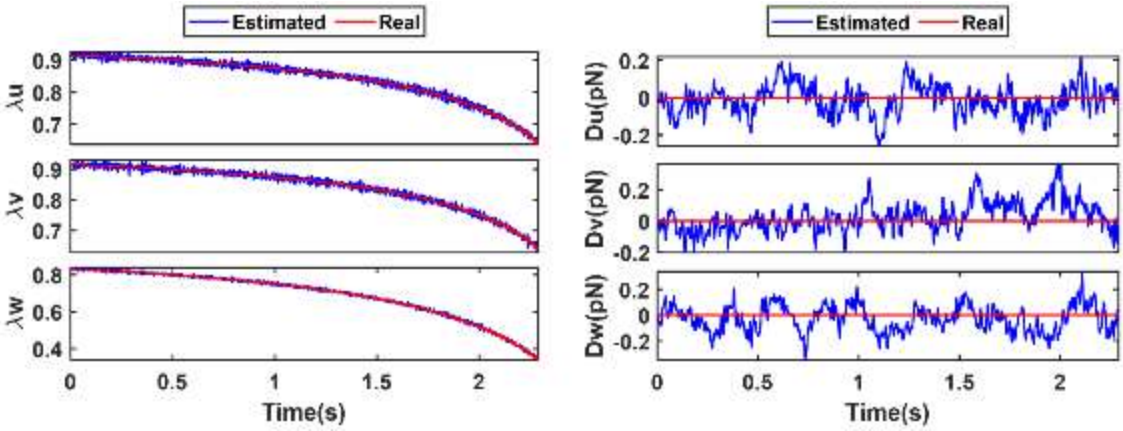

Fig. 42. The left panel shows the real-time estimation of the normalized input gain in the local coordinate, and the right panel shows the estimated disturbance.

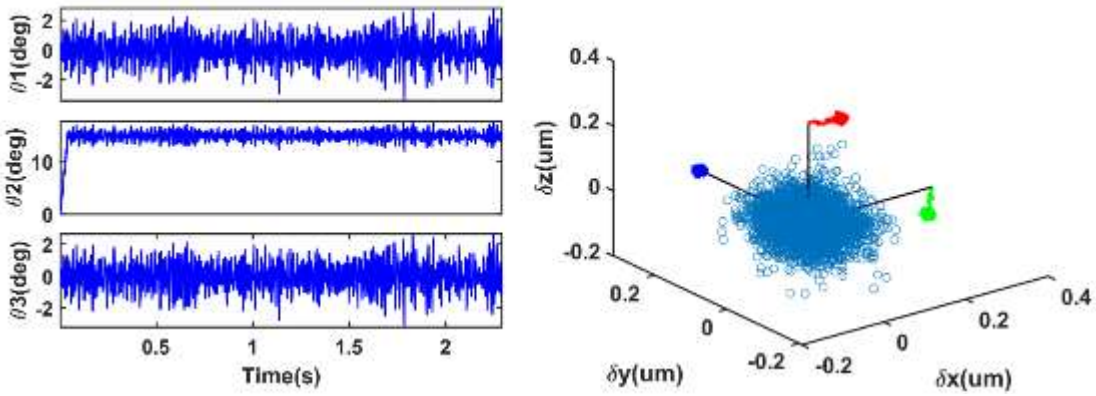

Fig. 43. The left panel shows the real-time estimation of the orientation of the surface normal, and the black line of the right panel is the positive direction of the measurement coordinate, while the dark blue, red, and green curve are the estimation result of the orientation of the surface normal in the measurement coordinate.

5.6 Conclusion

When operating in an unstructured environment, i.e., having inclined planes and/or spherical surfaces, although the drag coefficient can be characterized by two main directions, perpendicular and parallel to the surface, the surface orientation is unknown. The three separate drag coefficients will turn into a 3×3 drag coefficient matrix in the measurement coordinate depending on the orientation of the surface normal.

It would be complicated if we directly estimate the 3×3 drag coefficient matrix. Instead, we estimated the orientation of the surface, utilizing the covariance matrix of the tracking error to predict and estimate the orientation of the surface through real-time principal component analysis [60]. Since the covariance matrix is positive definite

symmetric matrix, it can be written in the form of eigenvectors which are characterized by three angles and diagonalized eigenvalues.

By multiplying the eigenvector matrix, we can transform the system from the measurement coordinate to the local coordinate. The 3×3 input matrix of the scanning microprobe in the measurement coordinate is diagonalized in the local coordinate. The control law and augmented estimator can be derived similarly to the method discussed in Chapter 4. The estimation of the orientation of the surface is also included in the augmented estimator.

Two experiments were conducted to verify the control and estimation of the scanning microprobe in the designed scenario. We can correctly estimate the orientation of the tilted coverslip no matter if we rotate the coverslip in the x or y axis. The tracking error in the local coordinate is zero-mean. The estimated normalized input gain followed the real input gain tightly, and the estimated disturbance is subjected to the thermal force and measurement noise. This lays a foundation for the ability of scanning and indenting 3-D objects in the workspace.